Mainak Deb

Neuro-AI PhD student @ Georgia Tech

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Research

October 2026 **Braintree: An agentic framework for in-silico vision neuroscience**
Mainak Deb, Ruolin Wang, Alish Dipani, Apurva Ratan Murty
Under review, ICLR 2027. bioRxiv preprint.

April 2026 **Model-derived Minimal Image Pairs Predict Differential Responses in Human Scene-Selective Cortex**
Junxia Wang, **Mainak Deb**, Haider Al-Tahan, Diego Garcia Cerdas, Iris Groen, Apurva Ratan Murty
Under review, CCN 2026

February 2025 **TopoNets: High performing vision and language models with brain-like topography**
Mayukh Deb, **Mainak Deb**, Apurva Ratan Murty.
ICLR spotlight 2025 (top 2%)

December 2024 **Small-scale adversarial perturbations expose differences between predictive encoding models of human fMRI responses**
*Nikolas McNeal, ***Mainak Deb**, Apurva Ratan Murty.
NeurIPS 2024 Unireps Workshop.

January 2024 **Microarchitectural Design Variation of the Echinoid Skeleton: A 3D Structural and Mechanical Study of Paracentrotud lividus**
Valentina Perricone, Pasquale Cesarano, **Mainak Deb**, Derek Lublin, Mirko Mutalipassi, Lucia Pappalardo, David Kisailus, Francesco Marmo.
(Preprint)

Industry

February 2025 **Research @ Tether Evo**
- Developed encoding models using pre-trained AI models to predict whole brain activity (fMRI) encoding multimodal input/stimuli (vision, audio and language). 15th Rank on algonauts 2025 challenge.

January 2024 **Research Intern @ MurtyLab (Dr. Ratan Murty), Georgia Tech**
- Developed projects dealing with adversarial robustness in models trained on brain data, inducing brain-like topography in ANNs, and simulating the effects of brain perturbations (lesion/stroke) on human visual perception.
- Leveraged fMRI datasets (NSD, DeepRecon, Murty185), built a python package to seamlessly interface between fMRI data to pytorch dataloader.

July 2022 **Research Engineer @ Artflow.ai**
- Built fine-tuning and prompting pipelines for GPT-3, GPT-3.5 and GPT-4.
- Used LLMs reasoning with text and audio embeddings to auto-retrieve voices, music, sound effects and shot types.
- Used 3D inpainting to create cinematic camera moves and depth-of-field effects from still images.

July 2021 **Google Summer of Code 2021 Intern @ International Neuroinformatics Coordinating Facility**
Trained deep neural networks from scratch to help map the embryogenesis process in C. elegans worm embryo and deployed models live on the web using ONNX. Feel free to check out the detailed report and work repository

January 2022 **Research Intern @ Hybrid Design Lab, University of Campania, Italy**
Worked on generating time-varying textures that emulate Sea Urchin skeletons using Neural Cellular Automata and experimented with the latent output channels (apart from RGB) to generate vibrant psychedelic patterns. This was showcased at the Echino Design exhibition, 24 February 2022 at Città della Scienza, Naples, Italy.

March 2022 **Research Intern @ Department of Structures for Engineering and Architecture of University of Naples Federico II**
Worked with Prof. Francesco Marmo (and team) on bio-mechanical research that has the goal of geometrically and mechanically characterizing the stereom of echinoid exoskeletons. I processed micro-CT data to extract geometrical models of the stereom using computer vision, here's the Github repository and corresponding preprint

Personal Projects

Pre-AI, purely to teach myself

December 2021 **Text-2-Neural Cellular Automata**
[Github, Blog post]
Generate beautiful cellular automata patterns from natural language prompts, using CLIP-guided Neural Cellular Automata, built using PyTorch.

December 2020 **Deceptive Digits**
[Github, Blog post]
Conditional GAN to generate hand written digit images.

July 2020 **Watch Your Step**
[Github, Notebook]
Tabular Q-learning agent that teaches itself to cross OpenAI Gym's Frozen Lake over 9,500 self-played episodes, reaching the goal even in the stochastic "slippery" setting where intended moves often fail.

May 2022 **Lovecraft-GPT**
[Github]
Implemented and trained a word-level GPT from scratch in PyTorch on H.P. Lovecraft's novels, generating text in the source's style.

November 2020 **Eyes on the Road**
Train a Pytorch CNN to classify driver activity (texting/talk on phone etc.), Test on my brother driving.

August 2020 **Bank me Later**
Predict Bank Term deposit subscription probability using attributes like job, marital status, age etc with an accuracy of 94%

March 2020 **Deep Wine Connoisseur**
Use a pytorch dense net to predict wine-quality from it's chemical attributes.

February 2020 **Facial Expression Classifier**
PyTorch CNN to classify facial expressions w/ real-time inference

Education

2024 – Present **PhD Student**, *MurtyLab*, Georgia Institute of Technology

2023 **Bachelor's of Technology**, *Electrical and Computer Engineering*, Amrita Vishwa Vidyapeetham

Technical Skills

Languages Python, Bash, LaTeX

ML / DL Recursive Self-Improvement (RSI), Agentic Systems, LLMs, PyTorch, Vision Transformers, CNNs, CLIP, GANs, Adversarial Attacks, Reinforcement Learning, ONNX

Neuroimaging fMRI Encoding Models, Pycortex, Freesurfer, Brain-Score, NSD

Data & Vis NumPy, SciPy, Pandas, OpenCV, PIL, Matplotlib, Seaborn, Plotly

Infra Slurm, HPC, Git